

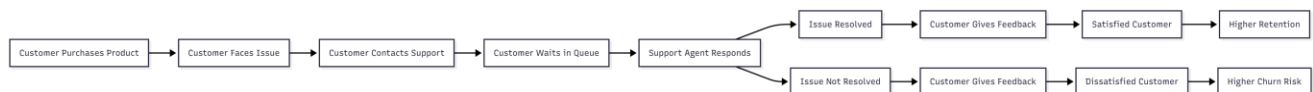
Customer Service Impact on Retention: A Call Data Study

1. Introduction

This analysis aims to evaluate how customer service performance influences customer retention at Flipkart. The study focuses specifically on identifying issues within customer support operations that may be contributing to reduced customer satisfaction and, ultimately, customer churn.

To achieve this, customer call data will be examined, along with key performance indicators such as Customer Satisfaction (CSAT) scores and service efficiency across different call centres and support channels. By understanding the relationship between service quality and customer retention, this project seeks to highlight operational gaps and provide insights to enhance Flipkart's customer service strategy. The overall objective is to support long-term customer retention by improving service delivery and customer experience.

2. User Journey Map



Customer Service User Journey – Bullet Points

- Customer purchases a product on Flipkart
- Customer receives product
- Customer faces an issue related to product/service/delivery
- Customer decides to contact Flipkart support
- Customer connects through support channel (call/chat/email)
- Customer waits in queue for assistance
- Support agent responds and listens to the issue
- Support works on resolving the problem

Two possible outcomes:

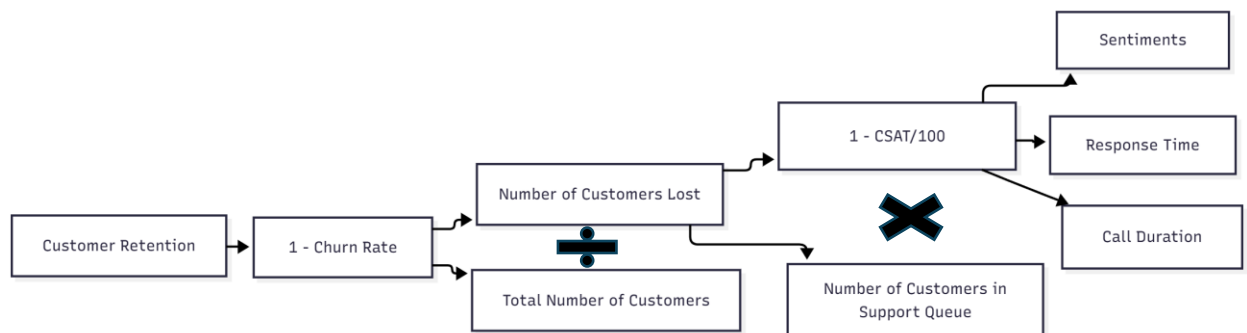
Issue Resolved

- Customer feels supported
- Customer provides positive feedback
- Customer satisfaction increases
- Customer is more likely to stay/return (higher retention)

Issue Not Resolved

- Customer feels dissatisfied and frustrated
- Customer still submits feedback (usually negative)
- Low satisfaction score and negative sentiment recorded
- Customer may stop using the platform (higher churn risk)

3. Metric Tree



Metric Tree Description

- The metric tree illustrates how customer service performance influences customer retention.
- Customer retention is determined by the inverse of churn rate, where churn rate depends on the proportion of customers lost to the total customer base.
- Customer loss is primarily linked to customer support experience and service quality.
- A key driver in this model is **Customer Satisfaction (CSAT)** — lower CSAT increases the likelihood of churn.
- CSAT is influenced by three core operational factors within customer service:
 - **Response Time** – how quickly customers are connected to support.
 - **Call Duration** – how efficiently customer issues are resolved.

- **Customer Sentiments** – emotional tone captured from customer feedback or interaction transcript analysis.
- Higher response time, long call duration, and negative sentiments reduce CSAT, which increases customer churn and decreases retention.
- This metric tree helps identify critical service factors that must be improved to strengthen customer retention at Flipkart.

4. Key Metrics to Track

To assess the impact of customer service on customer retention, the following key performance metrics are evaluated:

- **Customer Satisfaction Score (CSAT):**
Measures the customer's satisfaction level after their support interaction. Higher scores indicate positive customer experiences, while lower scores reflect dissatisfaction. CSAT is calculated based on customer feedback submitted post-interaction.
- **Customer Sentiment Analysis:**
Sentiments such as *very positive* and *positive* indicate satisfied customers, whereas *negative* and *very negative* sentiments suggest dissatisfaction. This metric provides qualitative insight into customer emotions and is further analysed at the call-centre level.
- **Response Time:**
Represents the time taken for a customer to connect with a support agent. When response time meets or stays below the Service Level Agreement (SLA) threshold, customers tend to remain satisfied. Delays beyond SLA may result in frustration and negative sentiment.
- **Call Duration:**
Refers to the time required to resolve a customer issue. Efficient and shorter resolution times generally indicate effective support and contribute to higher customer satisfaction. Longer call durations may signal inefficient processes or complex issues, potentially lowering satisfaction.
- **Percentage Distribution of Complaint Reasons:**
Examines the frequency of different types of customer complaints. A recurring pattern in similar complaint types suggests unresolved product or service issues, negatively impacting customer trust and increasing the risk of churn.

5. Data Cleaning

This section outlines the steps taken to prepare the dataset for analysis, ensuring accuracy, consistency, and usability.

1. Data Import

- The dataset, provided in CSV format, was imported into Microsoft Excel for initial review and processing.

2. Handling Missing Values

- The dataset was examined for missing entries across all fields.
- **CSAT Scores:**
- A significant number of missing values were found in the `csat_score` column, indicating that many customer interactions did not receive feedback.
- These values were retained, as removing or imputing them could distort analysis and they may hold meaning (e.g., customers not providing feedback due to dissatisfaction or neutral experience).
- **Other Fields:**
- Minor missing values were observed in fields such as `customer_name`, `gender`, `city`, and `state`.
- Since these fields were not critical to the core analysis, no imputation was applied. Missing values were simply noted.

3. Removing Duplicates

- The dataset was checked for duplicate rows to maintain data integrity.
- No duplicate entries were identified, so no removal was necessary.

4. Standardizing Data Formats

- **Call Timestamp:** The `call_timestamp` column was standardized to a consistent date-time format using Excel's number formatting options.
- **Call Duration:** The call duration field was converted to a numeric format to ensure uniformity and was reviewed to confirm all values represented minutes.

5. Data Processing

- New derived columns were created to enhance analysis:
- **Call Date:** Extracted from `call_timestamp` to allow date-based trend analysis.
- **Day of Week:** Generated from `call_timestamp` to study patterns across weekdays (e.g., Monday, Tuesday).

6. Data Manipulation

To make the analysis more accurate and structured, a few new data columns were created:

- **Day of the Week:** Derived from the call date using the `DAY()` function to identify if certain days show different call centre performance.
- **Call Date:** The original date column had inconsistent formats, so `LEFT()`, `RIGHT()`, `MID()`, and `DATE()` functions were used to standardize it across all records.

7. Data Analysis and Insights

Hypothesis 1: Certain call centres have consistently lower CSAT scores compared to others.

Key Findings: The analysis of average CSAT scores across call centres revealed noticeable variation. Kolkata recorded the lowest CSAT score, whereas Mumbai and Delhi maintained relatively higher satisfaction levels.

Conclusion: Kolkata's poor CSAT performance suggests inefficiencies in service quality or staffing. Focused training, improved workflow, and better resource allocation are recommended.

Hypothesis 2: Call centres with higher response times tend to have lower CSAT scores.

Key Findings: Correlation between response time and CSAT was weak and statistically insignificant. Despite higher response times in some centres, satisfaction did not consistently drop.

Conclusion: Response time is not a major factor affecting CSAT. Communication clarity and issue resolution quality have greater influence on customer satisfaction.

Hypothesis 3: Call centres with higher call durations have lower CSAT scores.

Key Findings: Analysis showed a weak negative correlation between call duration and CSAT. Longer calls did not necessarily reduce satisfaction, indicating customers value resolution more than brevity.

Conclusion: Call duration has limited effect on satisfaction. Emphasis should be on problem resolution and service effectiveness rather than reducing call length.

Hypothesis 4: Customers from certain locations report more issues, leading to lower CSAT scores.

Key Findings: Delhi and Mumbai had the highest complaint volumes, particularly in billing and service outage categories. These cities also showed slightly lower CSAT averages.

Conclusion: High complaint counts in major cities suggest operational strain or higher service demand. Strengthening regional service efficiency can improve CSAT.

Hypothesis 5: Locations with high issue reporting cause increased pressure on call centres, resulting in poorer service quality.

Key Findings: Regions with large customer bases such as Delhi and Mumbai corresponded with higher workloads and lower CSAT scores. Overburdened teams likely reduce service quality.

Conclusion: Redistributing workload or increasing staff strength in busy locations can help balance demand and maintain service quality.

Hypothesis 6: Specific dates experience more reported issues, impacting CSAT scores.

Key Findings: Weekend dates, especially Saturdays, showed higher complaint volumes and slightly lower CSAT scores. Peaks in issue reporting corresponded with drops in satisfaction.

Conclusion: High weekend workload negatively affects satisfaction. Additional staffing and workload forecasting are recommended for weekends.

Hypothesis 7: The day of the week influences CSAT due to varying service demand and availability.

Key Findings: Saturdays and Mondays had the most complaints, while midweek days were relatively stable. CSAT fluctuated accordingly.

Conclusion: Weekday patterns affect satisfaction. Scheduling staff based on daily demand trends can improve response times and CSAT.

Hypothesis 8: Certain issue types are associated with lower CSAT scores.

Key Findings: “Billing Question” accounted for about 72% of all complaints and had the lowest average CSAT. Other issues such as service outages had relatively higher scores.

Conclusion: Billing-related concerns are the primary driver of dissatisfaction. Improving billing transparency and communication can boost satisfaction.

Hypothesis 9: Some support channels are more effective at resolving issues, leading to higher CSAT scores.

Key Findings: Phone and web channels achieved better satisfaction compared to chatbots and email. Automated channels struggled with complex issue handling.

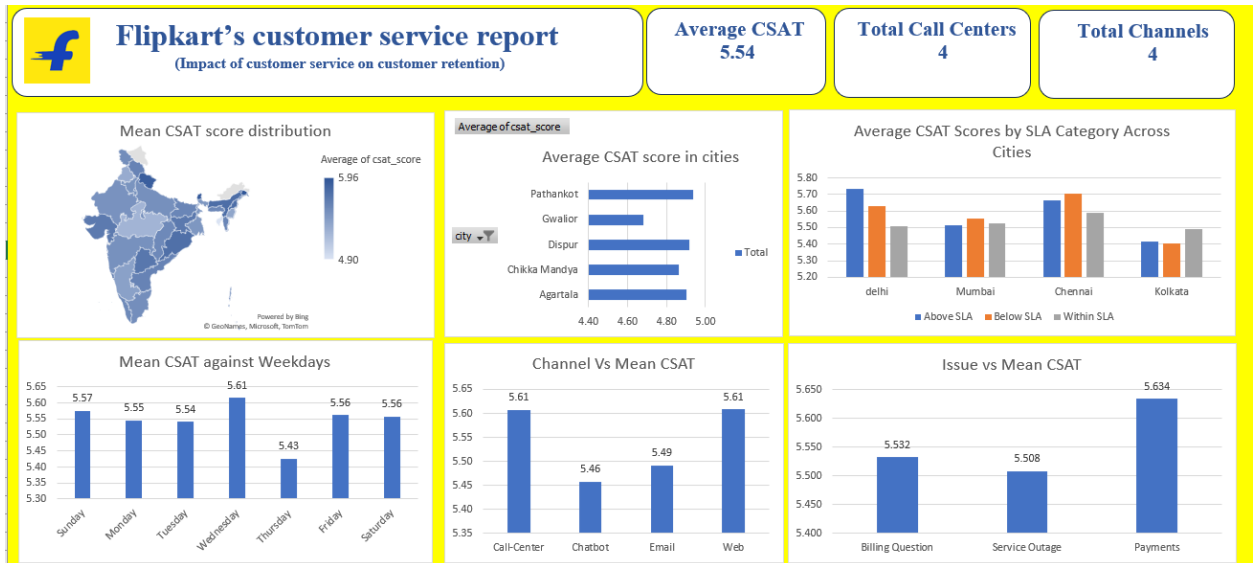
Conclusion: Enhancing chatbot capabilities and providing faster escalation to human agents can raise CSAT across digital channels.

Hypothesis 10: The effectiveness of support channels varies by call centre, affecting overall CSAT.

Key Findings: Channel performance differed by location — chat performed well in Mumbai but poorly in Delhi, showing inconsistent service standards.

Conclusion: Variation in channel performance highlights process inconsistency. Standardising service procedures across all centres will improve customer experience and satisfaction.

8. Data Visualization



9. Recommendation

- **Improve Underperforming Call Centres**

Focus on centres like **Kolkata**, which showed the lowest CSAT scores.

Implement targeted **training programs**, process audits, and quality monitoring to improve consistency in customer interactions.

- **Optimize Workload Distribution**

High-volume centres such as **Delhi** and **Mumbai** handle excessive complaints, affecting quality. Introduce **workload balancing** by redistributing calls, adding staff, or introducing regional support hubs.

- **Enhance Billing and Payment Systems**

Since most complaints relate to **billing issues**, review and streamline billing workflows. Improve **system accuracy**, simplify communication about charges, and introduce automated alerts for transaction confirmations.

- **Strengthen Support Channels**

Chatbot and **email** channels require refinement to match the performance of call and web support. Enhance chatbot training datasets, enable quick **human escalation**, and monitor satisfaction metrics per channel.

- **Adaptive Resource Scheduling**

Adjust staffing based on **daily and weekend patterns**. Increase active agents on **Saturdays and Mondays**, when issue volumes are highest, to prevent service delays and maintain consistent CSAT levels.

- **Standardize Service Quality Across Regions**

Develop **uniform service guidelines** for all call centres and channels to

eliminate inconsistencies. Use centralized training, standardized response templates, and periodic audits to maintain quality control.

- **Customer Feedback Loop Integration**

Incorporate **real-time feedback collection** after every resolved ticket. Analyze recurring dissatisfaction reasons monthly and take proactive corrective measures.

- **Performance Monitoring Dashboard**

Implement an internal **CSAT and Issue Tracker Dashboard** showing metrics by city, call centre, issue type, and channel. Continuous visibility will allow quick detection of drops in performance.

- **Root Cause Analysis for Low-CSAT Cities**

Conduct a deeper investigation into regions like **Gwalior**, which reported the lowest satisfaction, to identify hidden issues such as system latency, language barriers, or unresolved escalations.

- **Continuous Improvement Strategy**

Set up a quarterly review mechanism to evaluate trends in CSAT, issue categories, and channel effectiveness. Use these insights to continuously refine operational strategies and maintain long-term service excellence.